

Exercise 8:

```
lowerlim = [-4 0 -1];
upperlim = [4 6 3];
a=monte_carlo_gen(@integrand,lowerlim,upperlim,3,1000);
b=monte_carlo_gen(@integrand,lowerlim,upperlim,3,100000);
result = -448;
pea=100*abs((result-a)/result);
peb=100*abs((result-b)/result);
fprintf("n = 1000 Monte Carlo result = %f with percent error %f \n" + ...
        "n=100000 Monte Carlo result = %f with percent error %f ",a,pea,b,peb)
```

```
n = 1000 Monte Carlo result = -417.469088 with percent error 6.814936
n=100000 Monte Carlo result = -461.561339 with percent error 3.027085
```

We can observe that the percentage error of $n=100000$ is smaller than the percentage error of $n=1000$. Therefore our results are more accurate when n is big.

```
function [v] = integrand(x)
v = x(3,:).*(x(1,:).^3-2*x(1,:).*x(2,)-x(3,:));
end

function [I] = monte_carlo_gen(f,lowerlimits,upperlimits,IntegralDimension,n)
x=zeros(IntegralDimension,n); % initialize sampling matrix
mu=prod(upperlimits-lowerlimits); % calculate total volume
y=rand(IntegralDimension,n); % random numbers within [0,1]
% convert to within [a,b]
for i=1:IntegralDimension
    x(i,:)=lowerlimits(i)+(upperlimits(i)-lowerlimits(i))*y(i,:);
end
I=mu*sum(feval(f,x))/n; % evaluate at all points at once
end
```